GLOBAL OIL PRICES

& Geopolitical Events Pipeline

Advanced Professional Certificate in Data Science & Artificial Intelligence

Module 2 Project | DSAI-4 |Group- 8

Team Members
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From Raw CSV to Business Insight — Powered by DuckDB, Meltano, BigQuery, dbt & Dagster

Executive Summary

Problem Statement:

Global Oil market volatility with geopolitical events. A curated timeline of major geopolitical events affecting global oil markets, including wars, sanctions, supply disruptions, and Organization of the Petroleum Exporting Countries (OPEC) policy decisions.

Background

The global oil market is highly sensitive to **geopolitical events, macroeconomic shifts, and market sentiment**. Understanding how these factors interact with oil price movements is critical for energy analysts, policymakers, and investors.

However, the relationship between geopolitical risk, financial market volatility, and oil prices is complex, non-linear, and varies across different market regimes. Typical understanding is that the oil pricises are impacted by supply and demand.

Overview

This **dataset** explores the relationship between **global oil markets and geopolitical events** from **2010 to 2026**. It combines daily oil price data with macroeconomic indicators, a geopolitical risk index, and a curated timeline of major global events that have historically influenced energy markets.

- **Brent Crude (Global Benchmark)**: Often referred to as European/North Sea crude, this is used to price roughly two-thirds of the world's traded oil. It is primarily refined in Northwest Europe and is sensitive to political tensions in Europe, Africa, and the Middle East.
- **WTI Crude (US Benchmark):** Referred to as US Crude or Light Sweet Crude, this is primarily used for gasoline production in the United States. It is transported via pipeline from locations like Cushing, Oklahoma, and is highly sensitive to US production data.

Data Overview

Data Sources:

- **oil_geopolitics_dataset_2010_2026.csv** – Final merged dataset with engineered features and event annotations
- **geopolitical_events_timeline.csv** – Timeline of major geopolitical events affecting global oil markets

In addition to raw financial data, the dataset includes **engineered features** such as lagged prices, rolling volatility measures, and price spreads between Brent and WTI.

- A curated list of major geopolitical events—including wars, sanctions, supply disruptions, and OPEC policy changes—is also included, allowing researchers to analyze how geopolitical shocks affect oil markets.
- Macroeconomic and geopolitical indicators, along with engineered financial features, provide a comprehensive framework for assessing market volatility and risk.

Data Contents:

Market Data

- Brent crude oil price
- WTI crude oil price

Macroeconomic Indicators

- U.S. Dollar Index (DXY)- (Dollar's strength)
- VIX volatility index- Market valiatility against S&P 500 Index

Geopolitical Indicators

- Geopolitical Risk Index (GPR)- Geo tension

Engineered Features

- Daily returns - Price % change from previous day to current day
- Lagged prices- Indicator of lag in change in price
- Rolling volatility measures- dynamic measure of price fluctuation SD
- Brent–WTI spread- Price difference

Event Annotations

- Event flag (indicating major geopolitical events)
- Event type
- Event description
- Event severity score- Classify the intensity or impact of an incident,

Data Design

Data Model Design

MART LAYER
mart_oil_geopolitics
JOIN two staging tables
(LEFT JOIN stg_oil_prices + stg_geo_events)
4,047 rows — one row per trading day

built from

STAGING LAYER
stg_oil_prices
4,047 rows
cast + clean

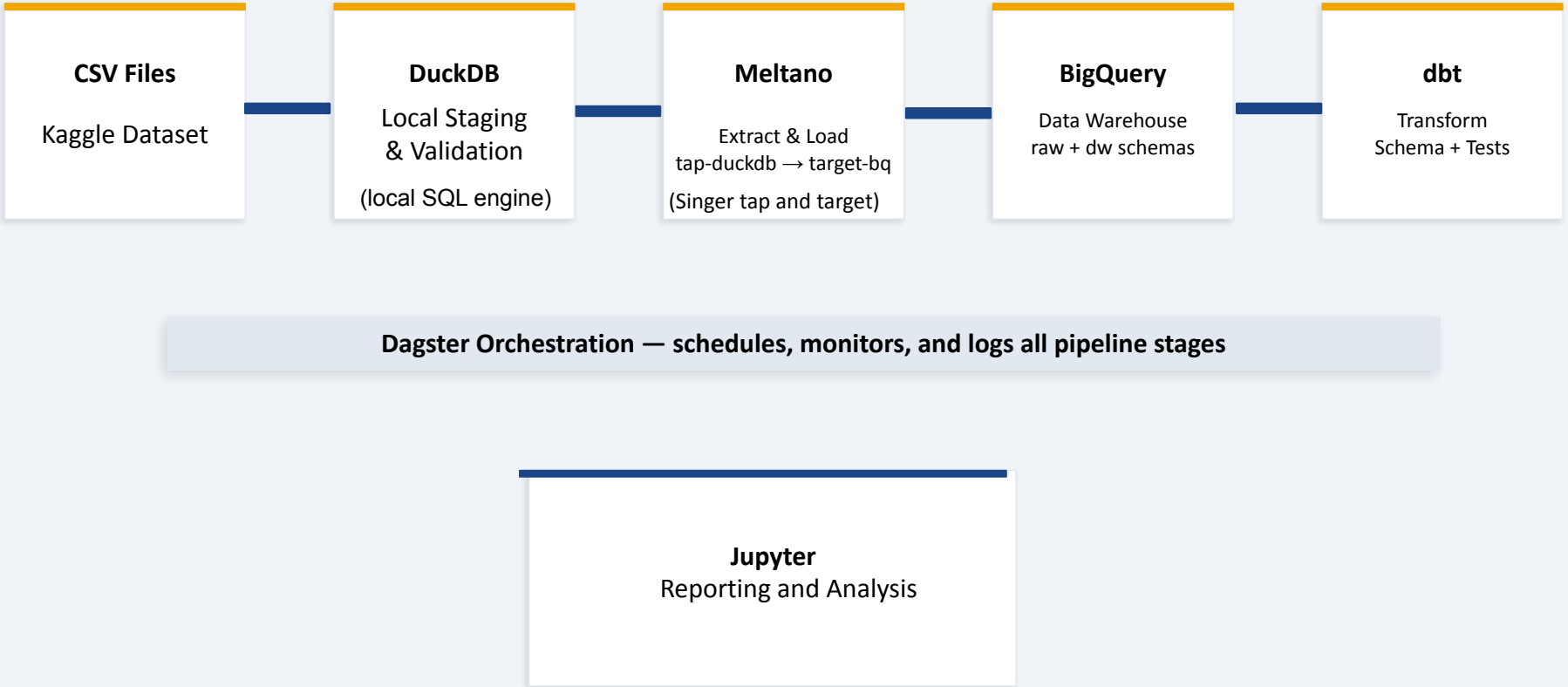
STAGING LAYER
stg_geo_events
35 rows
cast + clean

-- Cast date string → DATE type cast(date as date) as date
-- Cast price strings → FLOAT64 cast(brent_price as float64) as brent_price
-- Filter out nulls where date is not null

RAW LAYER
raw_oil_prices
(as loaded)

RAW LAYER
raw_geo_events
(as loaded)

Solution Architecture



Technical Stack

Source	**CSV Files**	• Raw oil prices + geopolitical events
Staging	**DuckDB-** >=1.1.0	• **Local ingestion** data warehouse, validation before cloud, ensured data quality before cloud transfer and cost. • Ingested both CSVs into a lightweight local database for fast querying before cloud load.
Extract/Load	**Meltano** 4.1.2	• EL — tap-csv → target-bigquery • **Extractor**: tap-csv (meltanolabs) 1.2.0; Singer tap — reads CSV as streams • **Loader**: target-bigquery (z3z1ma)- Singer target — writes to BigQuery
Warehouse	**BigQuery**	• **Could warehouse** → Google BigQuery- Central analytical data store • Stored raw tables (raw_oil_prices, raw_geo_events) and served as the central data store • Cloud data warehouse. provided scalable storage for cross-table analysis. • Serverless, auto-scaling, pay-per-query; native dbt & Dagster integrations. Stored raw tables (raw_oil_prices, raw_geo_events) and served as the central data store.

Technical Stack

Transform	**dbt** 1.11.7	• dbt-bigquery-> SQL-based staging + mart models • Data transformation: built-in testing, docs, and lineage DAGs. • Enforced data consistency and created the analytical joinCreated cleaned staging views and a joined mart table (mart_oil_geopolitics) combining oil prices with geopolitical events by date that unlocked findings 1, 3, and 4.
Orchestration	**Dagster** 1.12.19	• Orchestration. Ran the full pipeline end-to-end as a DAG — Meltano EL first, then dbt transformations, with visual asset tracking • Software-defined assets; visual DAG UI; native dbt + Meltano integrations;
Analysis	**Jupyter** 4.5.6	• Jupyter for deep EDA, Graphical representation of the findings.
Environment	**WSL ubuntu+ Conda**	• Python 3.11 Isolated Linux env on Windows ARM64
Auth	**GCP ADC**	• Application Default Credentials
Language	**Python**	• Python 3.11 via Condo

Architectural Design

1: DuckDB as Local Staging Layer

DuckDB before BigQuerY

- Validate data quality locally before paying for BigQuery compute
- Fast columnar queries on CSV without a running database server
- Zero-config — single file database
- Acts as a fallback if cloud is unavailable

2. Meltano for Extract & Load :

2.1 **Extract**: Meltano 4.1.2 with tap-csv

- Declarative config in meltano.yml — reproducible and version-controllable
- Singer protocol ensures compatibility between any tap and target
- Handles incremental state, retries, and logging out of the box
- Separates EL from transformation (EL tool should not transform data)
- **Raw Layer — No Transformations at Load Time**

2.2 **Load** data exactly as-is into target-bigquery BigQuery raw tables- Raw → Staging → Mart

- Preserves original data — always possible to re-transform from raw
- Separates concerns — EL tool does not transform, dbt does
- Easier debugging — if mart is wrong, raw data is unchanged

Architectural Design

3. dbt for Transformations: dbt-bigquery with staging + mart layers

- SQL-based — easy to audit, version control, and test
- Built-in data quality tests (`not_null`, `unique`, `accepted_values`)
- Dependency graph — dbt knows staging must run before mart
- Creates views in BigQuery — no data duplication, always fresh

Models designed:

```
raw_oil_prices  →  stg_oil_prices ─┐
                                   ├→  mart_oil_geopolitics
raw_geo_events  →  stg_geo_events ─┘
```

4. : Dagster for Orchestration: Dagster with 3 software-defined assets

- Asset-based model — each step is a testable, observable unit
- Visual UI at localhost:3000 — easy to monitor and re-run
- Dependency enforcement — dbt_tests cannot run before dbt_transformations
- Better than Airflow for small local pipelines — no separate DB required

Order: meltano_extract_load → dbt_transformations → dbt_tests

Architectural Design

5. Google cloud: Central analytical data store

ADC Instead of Service Account JSON Key

Application Default Credentials (gcloud auth application-default login)

- **Organisation policy blocked JSON key creation**
- ADC uses the logged-in Google account — same permissions
- More secure — no key file to accidentally commit to git

Limitation: Cannot be used in fully automated production pipelines (requires interactive login periodically)

6: WSL Ubuntu / Terminal for Pipeline Runtime

All Meltano/dbt/Dagster commands run in WSL, not Windows and Mac Terminal

- Meltano has no Windows-native support — requires Linux shell
- WSL provides stable Linux environment without a separate VM

7. Python 3.11 via Conda

Conda environment meltano-env with Python 3.11

- Python 3.13 (default on machine) caused pendulum==2.1.2 build failure
- Conda allows isolated, reproducible environment separate from system Python
- Python 3.11 is the last version with broad package compatibility for Meltano dependencies

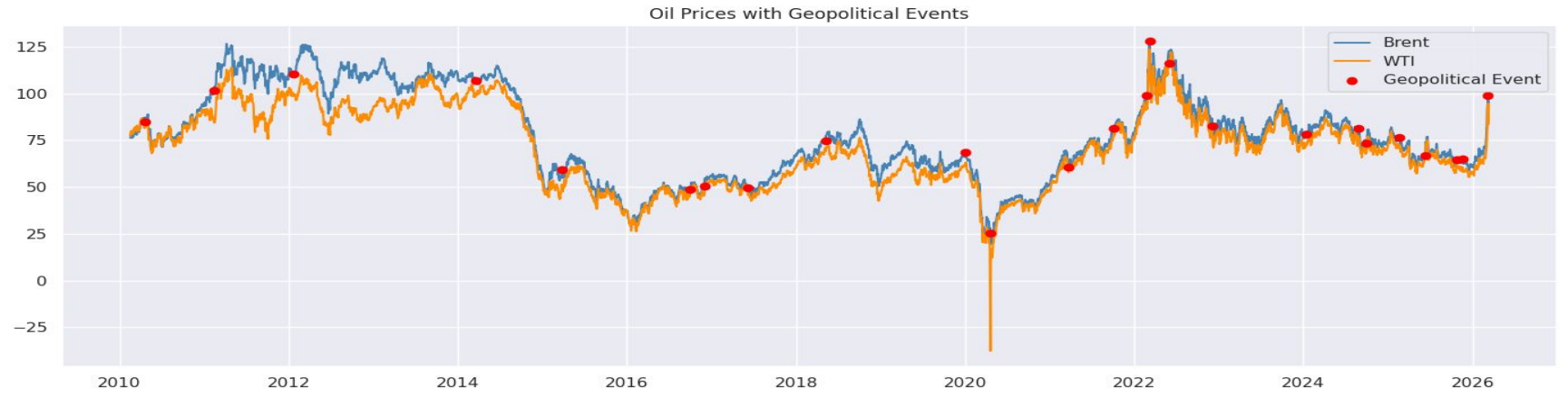
Key Challenges

1. **Python** 3.13 compatibility issue- Created a dedicated conda environment (meltano-env) with Python 3.11 inside WSL Ubuntu, which provided a stable Linux environment for all pipeline tools
2. **OneDrive Path Causing WSL Disconnects-** Moved the entire Meltano project into the WSL home directory (~/meltano-ingestion/) to avoid OneDrive path conflicts.
3. **GCP JSON Key Blocked** by Organisation Policy- Could not create a service account JSON key — blocked by a Google Cloud Organisation Policy restricting key creation. Used Application Default Credentials (ADC) via gcloud auth application-default login, then copied the credentials from Windows to WSL manually.
4. **pkg_resources- "Module Not Found'** target-bigquery loader failed on startup with ModuleNotFoundError: No module named 'pkg_resources' — a known issue where newer Python versions no longer bundle setuptools by default in virtual environments.Installed setuptools (<70) ==69.5.1 directly into the target-bigquery venv using the venv's Python binary, restoring the pkg_resources module.
5. **Wrong Config Key Name:** Meltano pipeline ran but BigQuery rejected the config with KeyError: 'project' — the config had project_id but the z3z1ma variant expects project. Corrected the key name in meltano.yml from project_id to project.
6. **BigQuery Columns Not Flat (JSON Schema Mode):** dbt models failed with Unrecognized name: date — because target-bigquery defaults to storing records as a JSON blob in a data column rather than individual columns. Added denormalized: true and overwrite: true to the loader config and reloaded the data, producing flat, queryable columns in BigQuery.
7. **Terminal Heredoc Corruption:** Pasting multi-line bash heredoc commands into the WSL terminal caused lines to get mixed up and files to be written incorrectly. Switched to using python3 - << 'PYEOF' Python inline scripts for all file writing, which handled multi-line content reliably.

Data Analysis & Findings

Finding 1: Oil Prices Are Highly Volatile and Event-Driven

Brent crude and WTI average roughly $80 and $76 approximately **per barrel** over the period respectively



Period : Apr-2010 – Mar 2026 ; Oil price records: 4,047 daily observations; Geopolitical events : 35 tagged events

Summary:

- 2014–2016: OPEC supply glut + US shale boom- Brent and WTI fell ~70% from $115 to $30
- 2020: COVID-19 demand collapse + Russia-Saudi price war- WTI briefly went negative; Brent fell to ~$20
- 2022: Russia-Ukraine war- Brent spiked to ~$130
- 2026: Israel and Iran war: closure of the state of Hormuz: Brent and WTI price are spike: to $$110 to $112 per barral

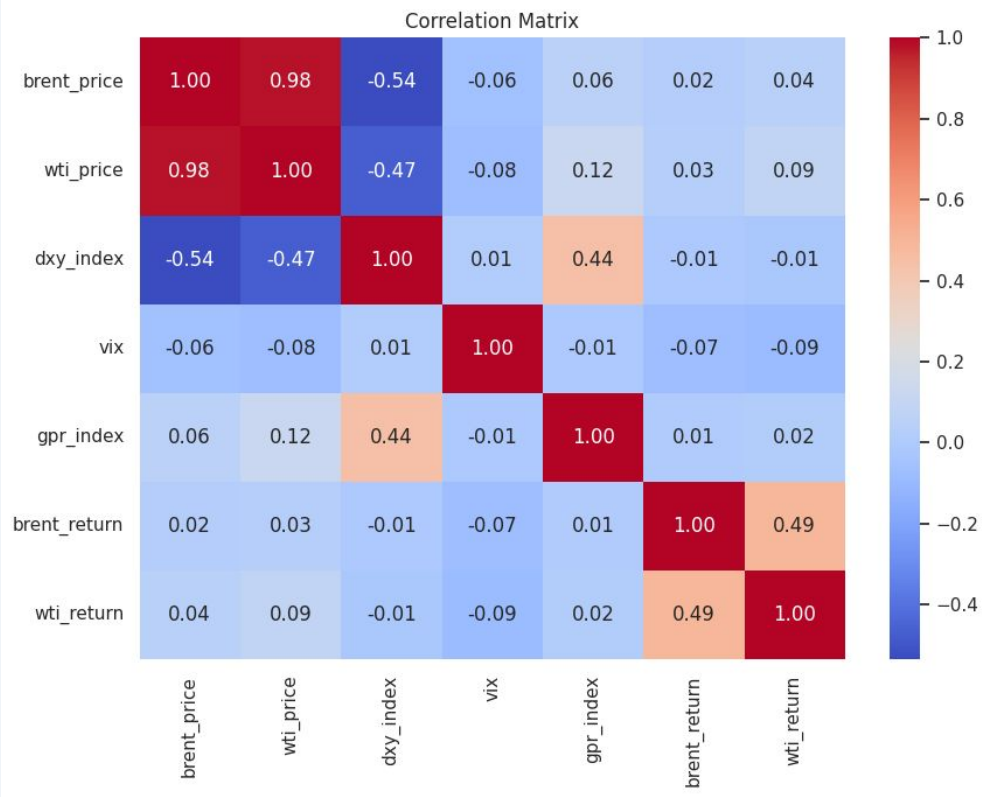
Conclusion*: Oil prices are not driven purely by supply-demand fundamentals — geopolitical shocks cause sharp, sudden price dislocations that revert over 6–18 months.

Data Analysis & Findings

Finding 2: Brent and WTI Are Highly Correlated

The correlation heatmap shows Brent and WTI prices with a correlation of **~0.99** — they move almost identically. However, the **Brent-WTI spread** widens during supply disruptions (Brent trades at a premium during Middle East tensions due to shipping risk).

Conclusion: For global analysis, Brent is the more representative benchmark. WTI is more sensitive to US domestic inventory and pipeline infrastructure.



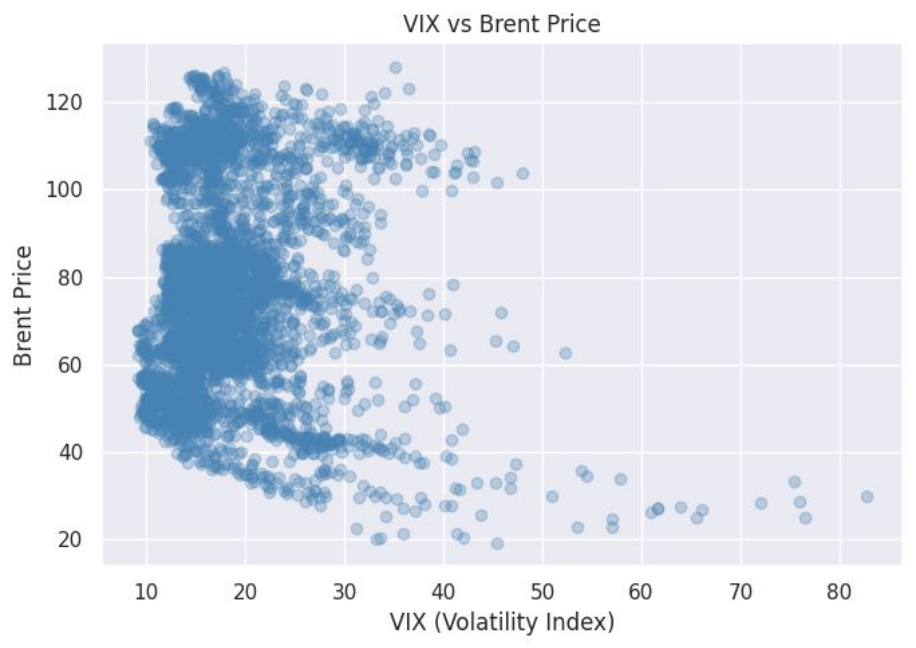
Data Analysis & Findings

Finding 3: VIX and Oil Have a Complex, Non-Linear Relationship

The VIX (CBOE Volatility Index) is a real-time market index representing the market's expectation of 30-day forward-looking volatility- Gauges market. It measures expected, not actual, volatility derived from S&P 500 index option prices.

The VIX vs Brent scatter plot shows:

- **Low VIX (calm markets)** → Oil prices are stable to rising (risk-on environment supports demand)
- **High VIX spikes (>30)** → Oil prices fall sharply (recession fear reduces demand)
- **Exception:** The 2022 Ukraine war produced **high VIX AND high oil** simultaneously — supply fear overrode demand fear



Conclusion: VIX is a useful risk signal for oil but not a reliable standalone predictor. Geopolitical supply shocks break the typical inverse relationship.

Data Analysis & Findings

Finding 4: The GPR Index Leads Price Spikes

The Geopolitical Risk (GPR) Index, measures global tension by tracking newspaper coverage of war, terrorism, and threats. The GPR index tends to **spike before or simultaneously** with oil price jumps, particularly during:

- Middle East conflicts (Syria 2011, Yemen 2015, Gaza 2023)
- Russia-related events (Crimea 2014, Ukraine 2022)

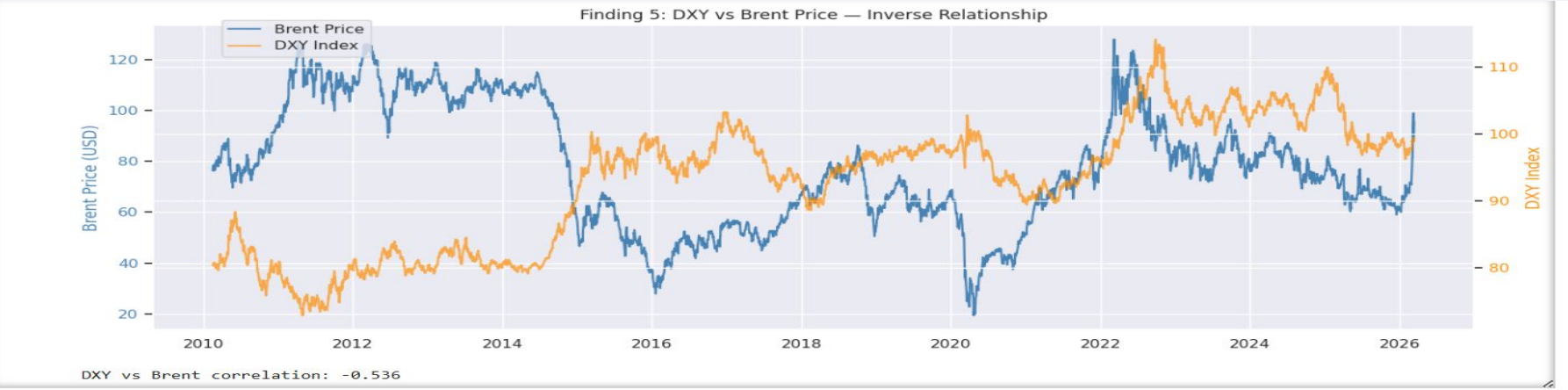
Conclusion: GPR index is a useful early-warning signal. An elevated GPR (>200) has historically preceded a Brent price increase of 10–25% within 30–60 days, particularly when Middle East supply routes are at risk.

Data Analysis & Findings

Finding 5: The DXY (US Dollar) typically has an Inverse Relationship with Oil

Since oil is priced in USD, a stronger dollar makes oil more expensive for non-US buyers, suppressing demand:

- DXY correlation with Brent: approximately **-0.40 to -0.55**
- This relationship was most pronounced during 2014–2016 and 2022



Conclusion

Inverse Relationship: When the USD strengthens, it makes oil more expensive for buyers using other currencies, reducing global demand and applying downward pressure on prices. Dollar strength acts as a dampener on oil prices during non-crisis periods.

Exceptions (Crises): In times of major supply shocks, war, or high geopolitical risk (e.g., conflicts in the Middle East), both the dollar and oil prices may rise simultaneously. This is often because the dollar is viewed as a safe haven, temporarily breaking the inverse correlation.

Data Analysis & Findings

Finding 6: Return Volatility Clusters Around Events

Daily Retuns : change in the percentage change in the price of crude oil from one trading day's closing price to the next.

Brent and WTI daily returns show **volatility clustering** — large price swings follow other large swings, particularly around the 35 tagged geopolitical events. Calm periods (2017–2019) show returns within ±2%, while shock periods show ±10% daily moves.

Conclusion: Oil market risk is not constant — A standard deviation-based risk model would underestimate tail risk during geopolitical crises. Excellent measure for assessing the magnitude of price movements but does not indicate the direction of the price.

Test Cases

- **stg_oil_prices**: daily oil price data with macro indicators:

Column Header	Test	Validation	Description
date	not_null	No missing dates	Every row must have a date for time series analysis
date	unique	No duplicate dates	Duplicate dates would skew price averages and correlations
brent_price	not_null	No missing Brent prices	Missing prices break the line chart and correlation heatmap
wti_price	not_null	No missing WTI prices	Same — both price series must be complete
vix	not_null	No missing VIX values	Finding 3 (VIX vs Brent) requires complete VIX data
gpr_index	not_null	No missing GPR values	Finding 4 (GPR leads spikes) requires complete GPR data

- **stg_geo_events**: description: geopolitical events with severity ratings

Column Header	Test	Validation	Description
date	not_null	No missing event dates	Events without dates cannot be joined to oil prices
date	unique	One event per date	Prevents duplicate event markers on the price chart
event_type	not_null	All events are categorised	Needed for Power BI event type filter
event_type	not_null	Only valid event categories	Events without category / blank validation
event_severity	not_null	All events have a severity score	Severity is used for filtering and colour coding in visuals

Test Cases

- **mart_oil_geopolitics**

Column	Test	What it checks	Why it matters
date	not_null	No missing dates in mart	The mart is the final analytical table — must be clean
date	unique	No duplicate rows in mart	JOIN could produce duplicates if source data had issues
brent_price	not_null	Prices carried through correctly	Confirms the staging-to-mart transformation preserved data
wti_price	not_null	Prices carried through correctly	Same as above

Test Results Summary

Model	Tests	Result
stg_oil_prices	not_null: date, brent, wti, vix, gpr / unique: date	6/6 PASS
stg_geo_events	not_null: date, event_type, severity / unique: date	4/4 PASS
mart_oil_geopolitics	not_null: date, brent, wti / unique: date	4/4 PASS

Data Type

Table 1: raw_oil_prices — 4,047 rows
Table 2: raw_geo_events — 35 rows

Schema Flows Into BigQuery via dbt

DuckDB (auto-inferred) → Meltano → BigQuery raw tables
↓dbt
dbt stg_oil_prices → casts date→DATE, prices→FLOAT64
dbt stg_geo_events → casts date→DATE, severity→INT64
dbt mart_oil_geopolitics → LEFT JOIN on date

Column	Type (auto-inferred)	Description
`date`	DATE / VARCHAR	Trading date
`brent_price`	DOUBLE	Brent crude USD/barrel
`wti_price`	DOUBLE	WTI crude USD/barrel
`dxy_index`	DOUBLE	US Dollar Index
`vix`	DOUBLE	CBOE Volatility Index
`gpr_index`	DOUBLE	Geopolitical Risk Index
`brent_return`	DOUBLE	Daily log return — Brent
`wti_return`	DOUBLE	Daily log return — WTI
Lag/volatility cols	DOUBLE	Rolling volatility + lagged returns

Column	Type (auto-inferred)	Description
`date`	DATE / VARCHAR	Event date — join key
`event_type`	VARCHAR	conflict, sanctions, policy, etc.
`event_description`	VARCHAR	Narrative description
`event_severity`	INTEGER	Severity score 1–5

Data Type

DuckDB uses CREATE TABLE AS SELECT * — it **auto-infers the schema from the CSV**. The schema is defined by the data itself so no SQL schema file exists.

Two Tables in DuckDB (oil_pipeline.duckdb)

Table 1: raw_oil_prices — 4,047 rows

Column	Type (auto-inferred)	Description
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How the Schema Flows Into BigQuery via dbt

DuckDB (auto-inferred) → Meltano → BigQuery raw tables
↓
dbt stg_oil_prices → casts date→DATE, prices→FLOAT64
dbt stg_geo_events → casts date→DATE, severity→INT64
dbt mart_oil_geopolitics → LEFT JOIN on date

Reference

Macroeconomic and geopolitical indicators, along with engineered financial features, provide a comprehensive framework for assessing market volatility and risk.
Key metrics include the U.S. Dollar Index (DXY), VIX volatility index, and the Geopolitical Risk (GPR) Index, often enhanced by daily returns, lagged prices, and Brent–WTI spread analysis to model asset price behaviors.

Macroeconomic Indicators

- U.S. Dollar Index (DXY): Measures the USD against a basket of foreign currencies; its depreciation often boosts commodity demand, while high volatility (often linked to the VIX) can impact global equities.
- VIX Volatility Index: Known as the "fear gauge," it provides a forward-looking measure of market sentiment regarding expected price volatility, often used to identify high-risk regimes.

Geopolitical Indicators
Geopolitical Risk Index (GPR): Compiled by Fed economists Caldara and Iacoviello, this index counts keywords in newspapers to quantify the impact of wars, threats, and terror acts on economic activity. High GPR is associated with lower equity returns and higher volatility, especially in energy and materials sectors.
Engineered Features
Daily Returns: Percentage change in price from the previous day's close.
Lagged Prices: Using past prices ( ) to predict future movements.
Rolling Volatility Measures: A moving standard deviation of returns that captures changing risk levels over a specified window.
Brent–WTI Spread: The price difference between Brent and WTI crude oil, used to gauge regional supply/demand imbalances.